Gregory Polyakov

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Education

PhD **University of Tübingen & Max Planck Research School**, Computer Science — Tübingen, Germany — Oct 2024 – present
- Health-NLP Lab, Supervisor: Prof. Dr. Carsten Eickhoff.
- Scholar, International Max Planck Research School for Intelligent Systems (IMPRS-IS).
- Research focus: Mechanistic interpretability of language models for NLP and IR tasks.

MSc **Moscow Institute of Physics and Technology**, Applied Mathematics and Computer Science — Moscow, Russia — Sept 2021 – July 2024
- Research topics: Sparse Retrieval, Multimodal Retrieval, Tool Calling in LLMs.
- Thesis: "Approaches to Training Language Models for External Function Calling: Using Self-Generated Data for Better Error Detection and Recovery".

BSc **Moscow Institute of Physics and Technology**, Applied Mathematics and Computer Science — Moscow, Russia — Sept 2017 – July 2021
- Thesis: "Usage of MCMC Methods in Deep Learning".

Experience

Huawei Noah's Ark Lab, NLP Research Engineer (full-time) — Moscow, Russia — Nov 2022 – Sept 2024 — 1 year 11 months
- Designed and co-implemented the end-to-end NLU pipeline for a production in-car voice assistant in Russian.
- Developed a fine-tuning method that teaches LLMs to detect and recover from failed API calls (ACL 2025 REALM Workshop).
- Co-developed a post-processing method for text-video retrieval (SIGIR 2023).

Huawei Noah's Ark Lab, NLP Research Intern (full-time) — Moscow, Russia — Oct 2021 – Nov 2022 — 1 year 2 months
- Co-developed the reranking model for a large-scale Russian-language product search engine: improved ranking quality by 5% relative NDCG@10 and cut inference latency 1.5x by adapting ColBERTer and TinyBERT-style distillation.

Publications

Understanding Wacky Weights: A Dissection of SPLADE's Learned Term Importance — 2026
Introduced metrics to formally quantify wacky weights (semantically unrelated expansion terms) in SPLADE sparse retrieval models. Showed that wacky weights mainly drive in-domain effectiveness, and measured which SPLADE settings promote more wacky weights.
G. Polyakov, H. Scells, C. Eickhoff
arxiv.org/abs/2605.19628 (SIGIR 2026)

Interpretability Analysis of Arithmetic In-Context Learning in Large Language Models — 2025
Identified the internal mechanisms LLMs use for in-context learning on arithmetic tasks. Used these findings to design a steering method that improves zero-shot performance, reducing the need for few-shot examples.
G. Polyakov, C. Hepting, C. Eickhoff, A. Bahrainian
aclanthology.org/2025.emnlp-main.92 (EMNLP 2025)

Towards Best Practices of Axiomatic Activation Patching in Information Retrieval 2025
Proposed a noise-mitigating evaluation metric for activation patching in neural rankers that filters out spurious low-confidence signals. Systematically analyzed diagnostic datasets, identifying key selection criteria for queries and documents to ensure robust interpretability analysis.
G. Polyakov, C. Chen, C. Eickhoff
dl.acm.org/doi/10.1145/3726302.3730256 (SIGIR 2025)

ToolReflection: Improving Large Language Models for Real-World API Calls with Self-Generated Data 2025
Proposed a fine-tuning strategy that enables LLMs to self-correct API calls by learning from runtime execution feedback. Co-developed evaluation datasets designed to assess tool-augmented LLMs in more realistic scenarios.
G. Polyakov, I. Alimova, D. Abulkhanov, I. Sedykh, A. Bout, S. Nikolenko, I. Piontkovskaya
aclanthology.org/2025.realm-1.14 (ACL 2025 REALM Workshop)

Sinkhorn Transformations for Single-Query Postprocessing in Text-Video Retrieval 2023
Co-developed a post-processing method based on Sinkhorn transformations that improves video retrieval quality. Introduced a realistic evaluation setting to compare post-processing methods in scenarios closer to real-world applications.
K. Yakovlev, G. Polyakov, I. Alimova, A. V. Podolskiy, A. Bout, S. I. Nikolenko, I. Piontkovskaya
arxiv.org/abs/2311.08143 (SIGIR 2023)

Scholarships

Presidential Grant Scholarship Feb 2018
Feb 2018 – Aug 2021. One of 1,200 recipients nationwide; four-year stipend awarded for academic merit and competition results.

Teaching

Teaching Assistant: Introduction to AI University of Tübingen
Winter Semester 2026/27. Conducted hands-on tutorials and evaluated student assignments.

Teaching Assistant: Modern Search Engines University of Tübingen
Summer Semesters 2025 and 2026. Conducted hands-on tutorials, evaluated assignments, and mentored student projects.

Teaching Assistant: Introduction to Python Programming University of Tübingen
Winter Semesters 2024/25 and 2025/26. Conducted hands-on tutorials and evaluated student assignments.

Skills

Programming: Python, C/C++, Bash

ML & NLP: PyTorch, Hugging Face Transformers, PEFT, Scikit-learn, CatBoost

Interpretability: TransformerLens, causal mediation analysis, steering, probing, sparse autoencoders, cross-layer transcoders

IR: Pyserini, FAISS, sparse and dense retrieval, neural reranking

Infrastructure: SLURM, multi-GPU training, Git

Languages

Russian: Native

English: C1 (IELTS Band Score 7)